Name: _____

Reg. No: _____

Ghulam Ishaq Khan Institute of Engineering Sciences & Technology

Faculty of Computer Science and Engineering

AI407L Lab Mid Exam - Spring 2026

Instructor: Mr. Muhammad Naeem, M. Asadullah Amin Total Marks = 150

Course Code: **AI407L** **Program: BSAI**

Course Title: **AI Capstone Project Lab**

Date: _____

Course Instructor Signature: _____

Vetted Faculty Signature: _____

HOD Signature: _____

Instructions:

1. **Follow the Lab Manuals Strictly**

All implementations must strictly comply with the specifications provided in Lab 2, Lab 3, Lab 4, Lab 5 for Part A.

2. **Functional Implementation is Mandatory**

Your system must be executable and demonstrably functional. Hardcoded outputs, mocked responses, or simulated behavior without actual pipeline execution will not be accepted.

3. **Modular and Clean Code Structure**
4. **Proper Naming Conventions**

Use meaningful variable names, function names, and file names that reflect system purpose. Poor naming reduces clarity and may affect grading.

5. **Independent Work**

This is an individual assessment. Collaboration or code sharing is strictly prohibited.

6. **Professional Documentation**

A proper technical report must accompany your submission. Poor documentation, even with working code, will affect marks.

7. **Submission Discipline**
   - Late submissions will not be entertained.
   - Missing files or incomplete components will receive zero for that section.
8. **Clarity Over Complexity**

Overly complex but unstable systems will not score higher than clean, well-structured, fully functional implementations.

Part A:

You are required to continue the same domain/project you have been developing in previous labs. You must implement all technical requirements exactly as defined in the official Lab Manual. Failure to follow lab specifications will result in deduction of marks. **(40 Marks)**

Task 1: [CLO-1/ GA-4/P4] (Marks: 10)

Design and **implement** a complete Retrieval-Augmented Generation (RAG) pipeline for your project domain.
All requirements must strictly follow the specifications defined in Lab 2: Knowledge Engineering & Domain Grounding.

Task 2: [CLO-1/GA-4/P4] (Marks: 10)

Design and **implement** the complete reasoning workflow of your agent using LangGraph.
Your system must:

- Define an appropriate graph state structure.
- Implement an agent node responsible for reasoning.
- Implement a tool execution node.
- Develop project-specific tools using proper decorators and input validation.
- Implement conditional routing logic to control execution flow.

Your workflow must demonstrate a functioning ReAct loop (Reason + Act).
All implementations must comply fully with Lab 3: The Reasoning Loop requirements

Task 3: [CLO-2/GA-4/P4] (Marks: 10)

Extend your system into multi-agent architecture using LangGraph.
You must **implement** the following:

- Define at least two specialized agent personas.
- Restrict tool access according to agent roles.
- Implement proper state handover between agents.
- Design routing logic to manage inter-agent communication.
- Demonstrate collaborative task execution requiring cooperation between agents.

Your output must clearly reflect distinct agent responsibilities and transitions.
All work must comply with Lab 4: Multi-Agent Orchestration requirements.

Task 4: **[CLO-2/GA-4/P4]** **(Marks: 10)**

Implement Persistent Memory and Human-in-the-Loop (HITL).

You must:

- Integrate persistent state management using a check pointer.
- Enable session recovery using thread identifiers.
- Identify a high-risk action tool within your project.
- Implement a safety interruption mechanism before execution of that tool.
- Allow human approval or cancellation before proceeding.
- Demonstrate that a human can edit the agent's proposed action before execution.

All work must strictly follow the technical specifications provided in
Lab 5: State Management & Human-in-the-Loop.

Part B:

You are required to implement a standalone Model Context Protocol (MCP) pipeline as a separate task.
This task must NOT be integrated into your Part A project.
You must design and demonstrate MCP in an independent use-case of your choice.
You must strictly follow the MCP design principles.
Failure to properly demonstrate MCP structure, context separation, and tool exposure will result in deduction of marks. **Marks:30**

Task 1: **[CLO-1/ GA-4/P4]** **(Marks: 10)**

Design and **implement** an MCP Server that exposes structured tools to a client model.

You must:

- Create a standalone MCP server.
- Define at least two callable tools inside the server.
- Clearly define structured input and output schemas.
- Ensure tools are exposed via proper MCP protocol specification.
- Demonstrate that the model accesses tools through MCP rather than direct function calls.

The implementation must clearly separate:

- Model
- Context
- Tools
- Execution layer

Task 2: **[CLO-1/GA-4/P4]** **(Marks: 10)**

Design and **implement** the MCP client-side interaction pipeline.

You must:

- Establish connection between model and MCP server.
- Demonstrate context exchange using the MCP standard.
- Show tool discovery from the MCP server.
- Execute at least one tool via MCP protocol.
- Capture and display structured responses.

Your submission must clearly show:

- Tool registration
- Tool invocation
- Context passing
- Response handling

Task 3: **(Marks: 10)**

Write a structured technical comparison explaining:

- Why MCP is needed in production systems.
- Difference between:
  - ✓ Direct Tool Invocation
  - ✓ LangGraph-based orchestration
  - ✓ MCP-based modular exposure
- How MCP improves:
  - ✓ Security
  - ✓ Scalability
  - ✓ System abstraction
  - ✓ Separation of concerns

This must be written as a formal technical justification.

Rubrics		
Component	**Marks**	**Weightage**
Part A (Agent System)	40	8%
Part B (MCP Pipeline)	30	4%
Technical Report	20	2%
Viva / Oral /written	60	6%
Total	150	20%